

COINDESK RESEARCH NOTE

On-Chain Transactions: The Novelty of Crypto Assets

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INTRODUCTION

A transaction is defined in the English language as "an exchange or transfer of goods, services or funds." Normally, the transfer of financial assets, especially if they are of a large size or quantity, is overseen by a trusted third party, such as a bank, a financial broker or an exchange. Using cryptocurrencies, a transfer of funds is not overseen by any single person or institution but rather is cryptographically secured through a technology known as a blockchain.

Blockchains, which were conceived as an idea as early as 1991, found their first commercial application through Bitcoin. Using blockchain technology, Bitcoin enabled online payments sent directly from one party to another in the bitcoin (BTC) currency. This ability to send funds without the oversight of a centralized authority has the potential to disrupt a $2.7 trillion global payments industry. But it's not only the global payment industry that could be impacted by a growing adoption for cryptocurrencies.

Since the creation of Bitcoin in 2009, hundreds of new cryptocurrencies leveraging blockchain technology have been launched. One of the most successful of these is ether (ETH), which runs on the Ethereum blockchain and which enables transfers of funds as well as various digital goods and services. On Ethereum, users can build virtual assets of any nature, fungible or non-fungible, and send them to other users. These assets may be representative of a rare piece of artwork, access to a marketplace as a buyer or supplier, or a subscription to an exclusive newsletter.

As such, sending transactions using cryptocurrencies has the potential to disrupt not only payments but virtually any transfer of value sent electronically from one person to another. In this research note, we will discuss three different metrics that track and measure transaction activity using cryptocurrencies. They are:

- On-chain transaction count.
- Total value transferred.
- Velocity.

We will also examine how these methods are applied differently outside of the context of cryptocurrencies and what that says about the novelty of blockchain technology applied to payments.

This is not intended to be a mathematical paper. We will look at some simple formulas, but it is intended for those who are not statisticians. We aim to make the measurement and understanding of cryptocurrency transaction activity accessible to everyone, as well as to underscore its importance to crypto assets and their potential role in investment portfolios.

We will refer mainly to bitcoin, as it is the largest and most liquid cryptocurrency, and the most often used as a benchmark for the industry when comparing crypto assets to traditional assets. When we use bitcoin with lowercase, we are referring to the asset, or BTC. Bitcoin with uppercase refers to the blockchain or the protocol. Dollars throughout are U.S. dollars.

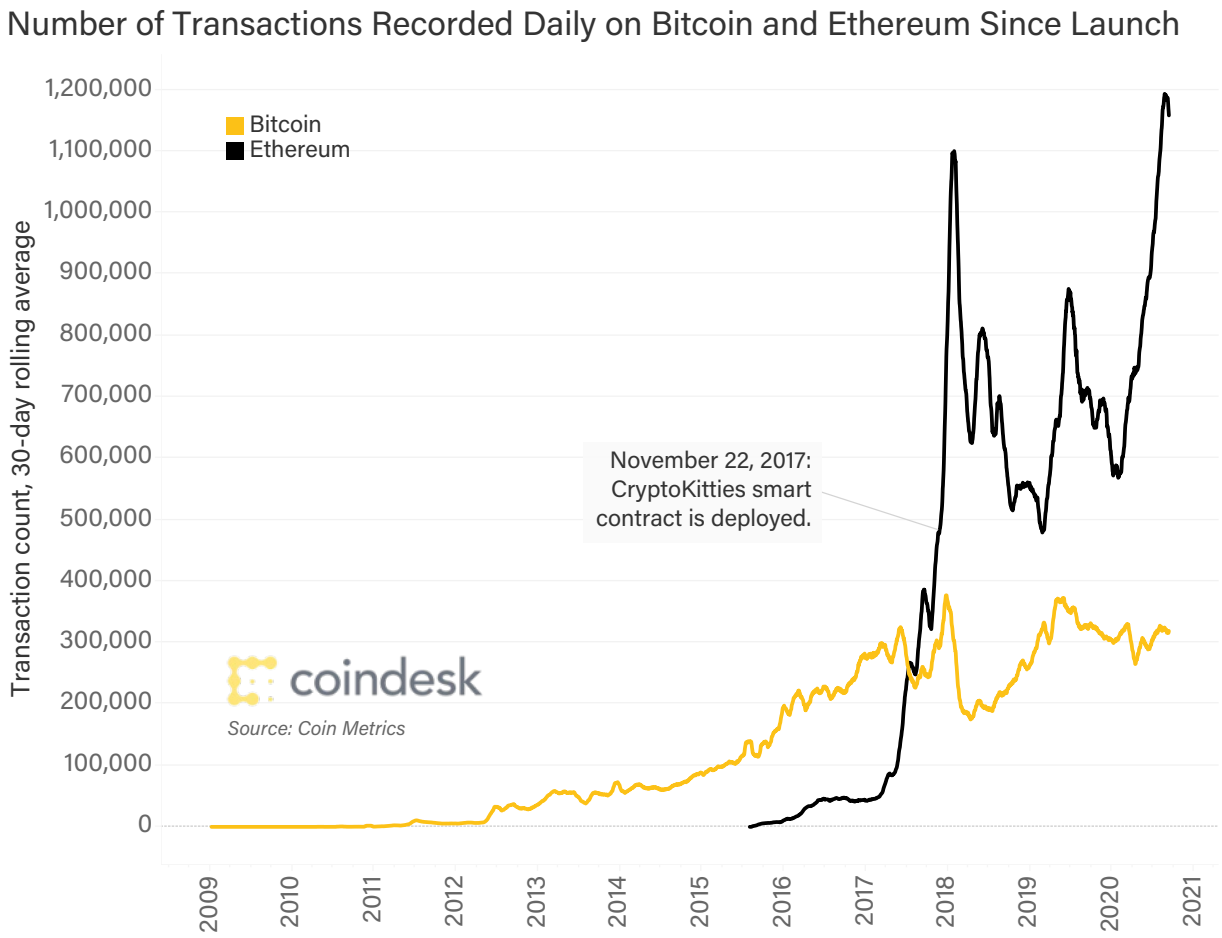
METRIC 1: ON-CHAIN TRANSACTION COUNT

The first and most basic metric measuring the transaction activity of a cryptocurrency is on-chain transaction count. Transactions processed "on-chain" are processed directly through a blockchain, meaning no-third party platforms are used to complete these value transfers. As such, they represent the activity of cryptocurrency users and businesses who have full custody over their crypto assets.

On Bitcoin, transaction activity from users with direct control over their BTC has surged over the past decade. From 2009 to 2018, the number of on-chain transactions processed daily grew exponentially from single digit numbers to over 375,000. While growth in this metric has since declined over the last 12 months, bitcoin's daily on-chain transaction count still regularly exceeds 300,000 (see chart, below).

On Ethereum, transaction activity has seen an even greater rise over the years, exceeding that of Bitcoin's consistently since late 2017. One of the key events in Ethereum's history that caused a surge in on-chain transaction count was the CryptoKitties decentralized application (dapp). CryptoKitties was the first dapp on Ethereum to go viral and attract a large user audience. It went live in late November 2018. Two months after its release, CryptoKitties reached a total of 250,000 registered users and generated $4.5 million on the Ethereum network.

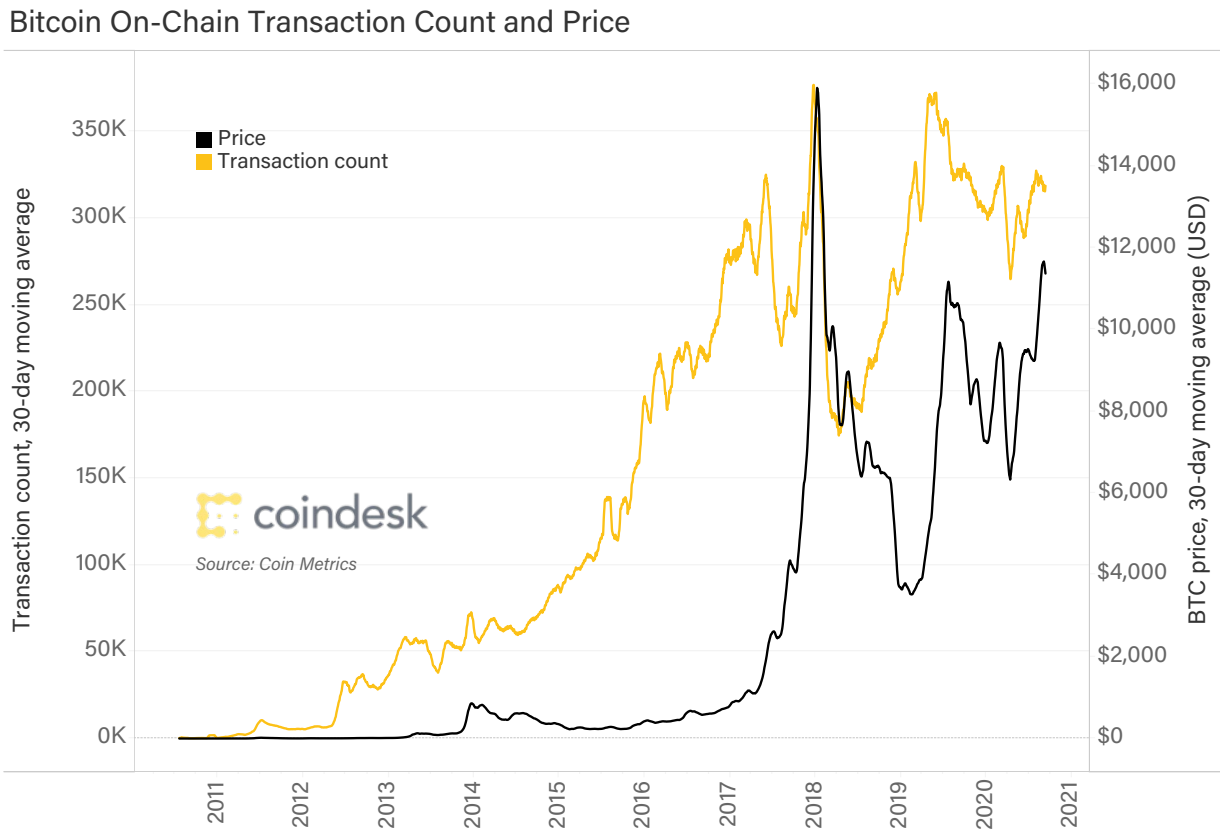
On-chain transaction count

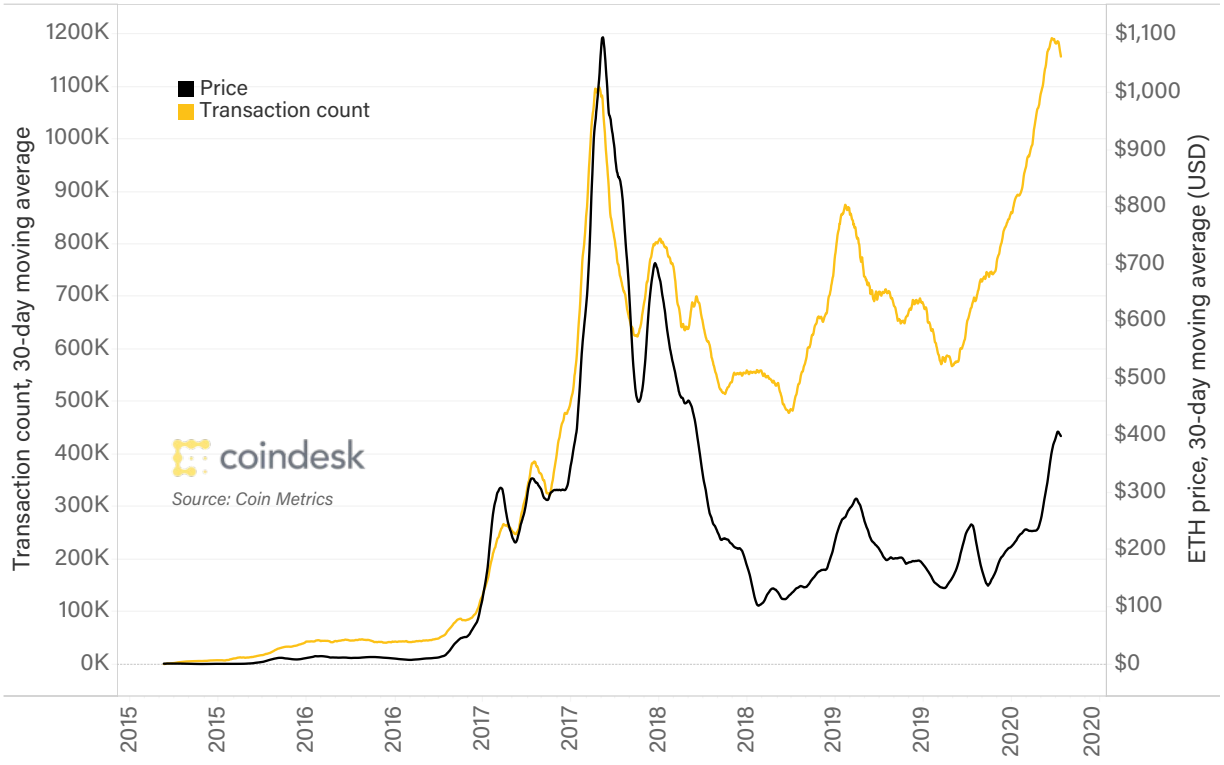


Comparing on-chain transaction count with the price of these two crypto assets over time reveals a close relationship between the two metrics. The highest recorded BTC price was reached at almost the same date Bitcoin recorded the highest number of on-chain transactions.

In the early years of Ethereum, ETH price and on-chain transaction count moved in almost perfect correlation. However, for both Bitcoin and Ethereum, the relationship between price and transaction count has weakened since 2018, with fluctuations in one metric not always leading to changes in the other.

On-chain transaction count and price





The apparent relationship between price and transaction count is likely to have compounding effects. A greater level of activity on a blockchain is a bullish signal for the platform's use and adoption that may cause investors to value it more highly in the markets. On the other hand, a higher price boosting the market value of a crypto asset could be taken as a vote of investor confidence in the network and further encourage its use by individuals and businesses.

However, there are a multitude of factors outside of on-chain transaction activity that weigh into a crypto asset's price that at times will cause prices to trend up even while the number of on-chain transactions goes down. These factors include but are not limited to trading activity happening off the blockchain, or "off-chain" for short. Large buys and sells for BTC on cryptocurrency exchange platforms like Coinbase or Binance could trigger subsequent trades of the same type in greater and greater volumes rippling across the market.

Regulatory action taken against crypto assets by national governments may also be one of these factors influencing changes in a crypto asset's price, as well as, on-chain transaction activity. It is difficult to identify all the various factors playing a role in this dynamic relationship between price and on-chain transaction count and also quantify the impact of ones occurring off-chain. However, as a starting point for analysis, on-chain transaction count is a relatively simple metric to understand and track.

METRIC 2: TOTAL VALUE TRANSFERRED

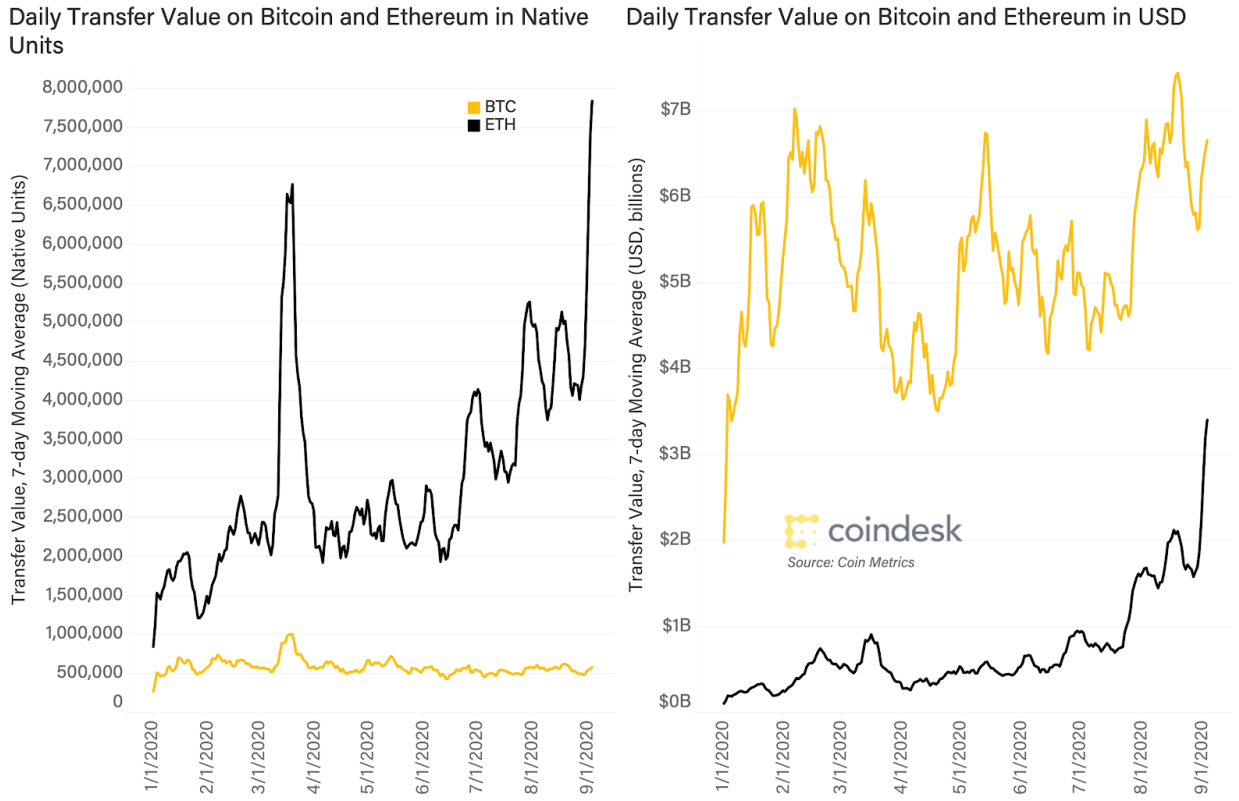
One example of a metric used to measure on-chain transaction activity that commonly requires off-chain data to calculate is total value transferred. This metric sums the value transferred through all transactions on a cryptocurrency network over a given period of time. Cryptocurrency networks can vary greatly in the number of transactions and the number of native units transferred, so comparing transaction activity in dollar terms can provide a better comparative measure of network activity.

Strictly using on-chain transaction data, total value transferred can only be summed in units of a crypto asset's native network currency. (This would be BTC for Bitcoin, ETH for Ethereum and so forth.) This is problematic for two reasons. First, cryptocurrencies are not widely used as units of account and hence, the values denominated in BTC or ETH for most readers would be arbitrary. Second, the values also would not be comparable across multiple crypto assets or traditional platforms.

To accurately weigh the total value transferred through one cryptocurrency vs. another network (crypto or otherwise) a common unit of account needs to be applied. This is most usually done by multiplying the market value of a crypto asset's native network currency (which is determined off-chain by exchanges) with the data gathered on-chain.

Applying this calculation to Bitcoin and Ethereum from January 1, 2020, we can see that while a larger volume of ETH is transacted on Ethereum than the volume of BTC on Bitcoin, in dollar terms the Bitcoin network consistently moves more value than Ethereum.

On-chain transaction data can be deceiving...



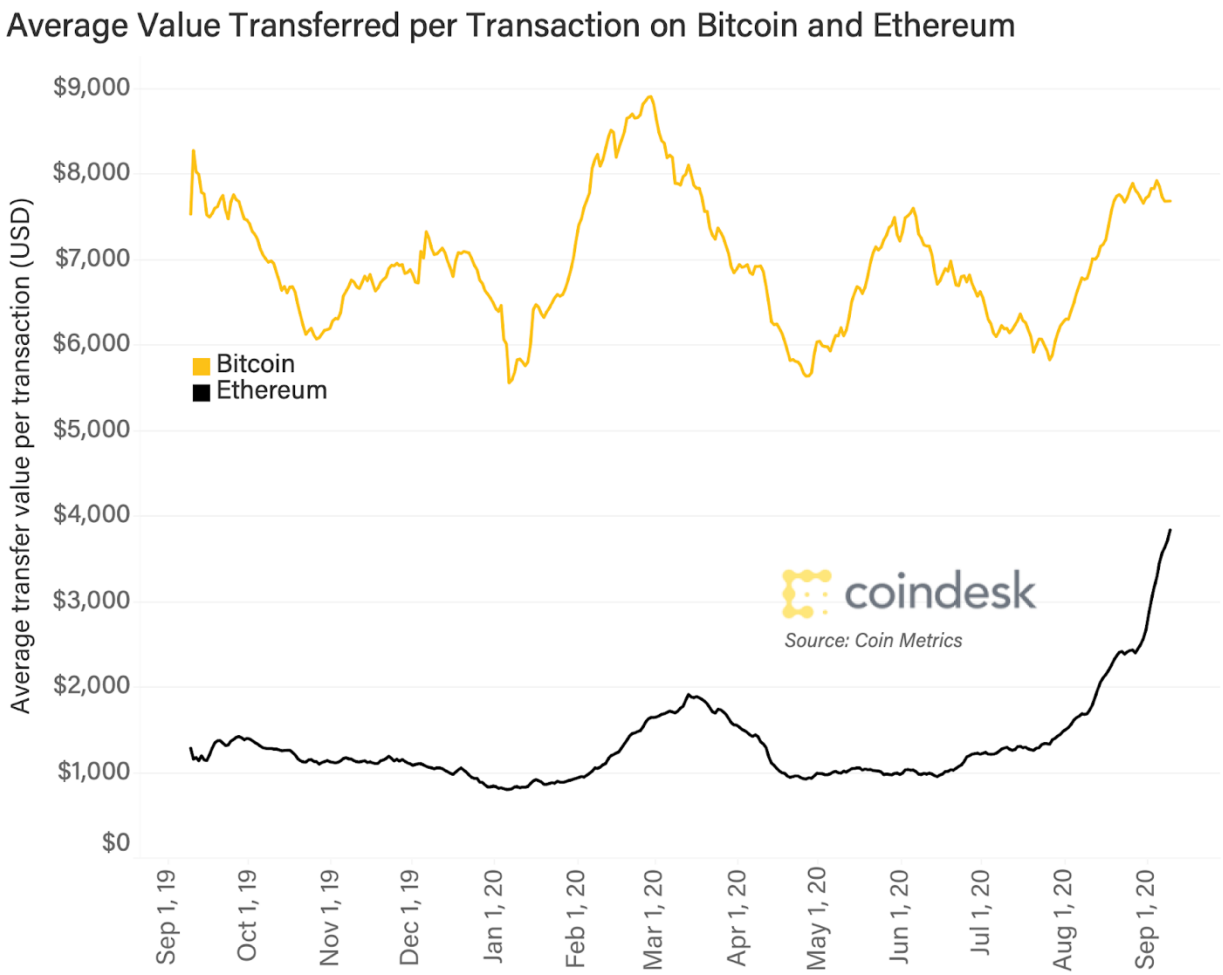
Total value transferred in dollar terms is not unlike another popular metric that is used to rank crypto assets. Rather than multiplying a crypto asset's market price with the amount of native tokens being transacted in the network, market capitalization multiplies an asset's price with its total circulating token supply. These values measure the relative size of a cryptocurrency within the broader crypto market.

Another related metric to total value transferred is the average value transferred per on-chain transaction. It is calculated by dividing the total value transferred across the network over a given period of time with the number of transactions that were processed in that same period.

Comparing the average value transferred on-chain between Bitcoin and Ethereum, it is clear that users on Bitcoin generally use the network to transact with a higher dollar value of BTC than users on Ethereum do with ETH (see chart, below). This observation strengthens the understanding that these two networks have different use cases.

Bitcoin is meant to be a secure and stable network for peer-to-peer payments. These payments are generally of high value, usually exceeding $6,000 on average. Ethereum is designed to be a network for deploying and interacting with decentralized applications (dapps). Rarely do we see users on Ethereum transferring large amounts of ETH in a single transaction to pay for using an application or executing one on the network.

Average value transferred can hint at differences in use case



The average transfer value per transaction on Ethereum, however, is rapidly increasing as of July 2020. While it is still only half the value of what is being transacted on average through Bitcoin, the trajectory of this metric for Ethereum is steep and positive.

One possible explanation for this trend is the recent surge of user activity generated by finance-focused dapps on Ethereum. Collectively called "Decentralized Finance" (DeFi), these types of dapps replicate the services of traditional financial institutions such as banks, lending platforms and exchanges. The nature of these services, more so than with gaming or gambling dapps, lends itself to attracting higher amounts of value and investment.

METRIC 3: VELOCITY

The final metric we'll look at for measuring the transaction activity of a cryptocurrency is velocity. In traditional economics, velocity is calculated by taking the ratio of a country's gross domestic product (GDP) to a country's money supply. The resulting value is a measure of how quickly money is being exchanged in the domestic economy.

A high velocity of money indicates money is changing hands frequently and, therefore, being spent more often than it is being saved. High money velocity is also taken as a sign by economists of a healthy and growing economy.

One of the first people to give a detailed overview on how velocity could be applied to understanding and valuing crypto assets is Chris Burniske, partner at crypto asset venture capital firm Placeholder Ventures.

In September 2017, Burniske published a book titled, "Cryptoassets: The Innovative Investor's Guide to Bitcoin and Beyond," which suggested viewing crypto assets as individual, isolated economies rather than private companies. Instead of revenues, margins and profits, Burniske suggested re-framing valuation models for cryptocurrency networks in the terms of GDP, size of monetary base and inflation.

Since Burniske's book, other prominent individuals in the cryptocurrency industry, including Ethereum founder Vitalik Buterin and founder of investment firm Multicoin Capital Kyle Samani, have made their own iterations on this model for evaluating crypto assets as self-governing economies.

For velocity calculations specifically, there are a number of different ways to measure the GDP and money supply of a crypto asset.

First, for GDP, the widest definition of this metric as applied to cryptocurrencies is total value transferred. Total value transferred represents the sum of all coins that have been moved from one address to another over a given period of time.

Dividing this by the total supply of coins at the end of the period illustrates how many times on average these coins were spent on a cryptocurrency network.

Velocity Calculation A:

GDP = Total value transferred

Money supply = Current total supply of coins

Velocity = SUM (Total value transferred) / Current total supply of coins

In addition, for money supply, taking the *average* of the total number of coins in circulation over a given time period is a slightly more accurate methodology for calculating velocity. For most

crypto assets with a fixed coin supply, the difference will not matter for the final results. However, for certain assets with a fluctuating supply that may differ month to month as a result of coins being burned (i.e., destroyed or made unspendable, a feature of some token economies), an average will ensure that any variations to coin issuance over a given time period are equally weighed.

Velocity Calculation B:

GDP = Total value transferred

Money supply = Current total supply of coins

Velocity = SUM (Total value transferred) / AVERAGE (Current total supply of coins)

Another common methodology for calculating velocity applies a narrower definition of the metric total value transferred. Instead of summing the values for all transactions occurring on-chain, adjustments can be made to exclude any transfers of value identified to be not representative of meaningful economic activity.

The line between what should be considered meaningful transaction activity and otherwise is a fine one. Some blockchain data companies have employed different techniques to better filter for the real economic throughput of a blockchain. One of these techniques removes on-chain transactions which are sent in rapid succession of each other in a short time interval. These transactions usually represent when a user is attempting to break down a large sum of value into smaller chunks and obfuscate where that value is going.

Velocity Calculation C:

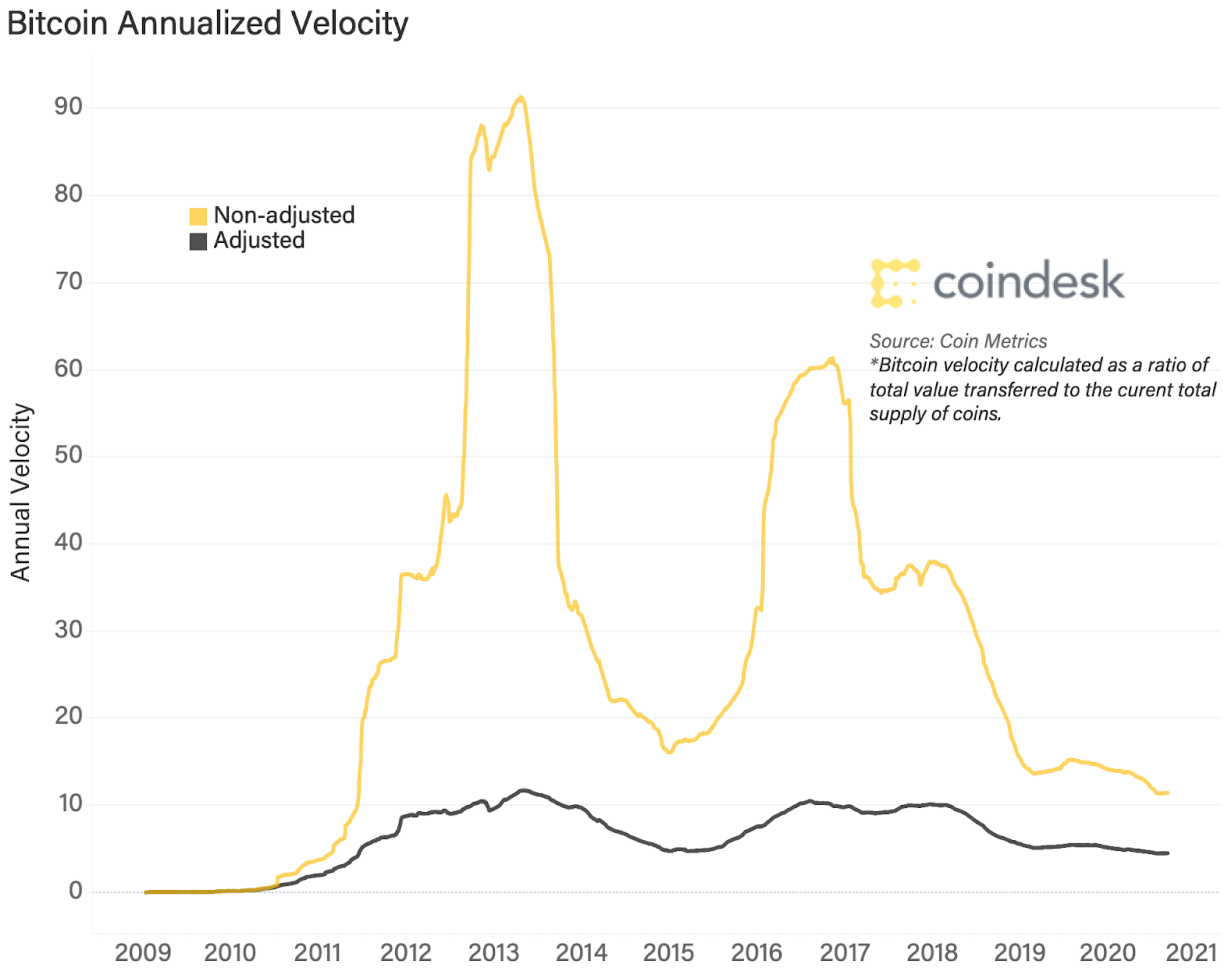
GDP = Adjusted total value transferred

Money supply = Current total supply of coins

Velocity = SUM (Adjusted total value transferred) / Current total supply of coins

Plotting velocity calculations A and C for bitcoin on the same chart illustrates a wide gap between using an adjusted measurement for total value transferred and the non-adjusted version. This suggests that a large fraction of on-chain transaction activity is not representative of meaningful economic activity.

A large fraction of on-chain transaction activity is not meaningful economic activity



While their values can differ greatly, the two calculations generally result in trends over time that rise and fall in tandem with the other.

As such, the velocity of a crypto asset plotted over time is illustrative of changes to how quickly coins and tokens are being spent as opposed to held on a cryptocurrency network. It is not, however, a direct representation of how frequently users of a crypto asset are transacting with their coins for the purchase of goods or services. This is because on-chain transaction data does not clearly differentiate between transactions initiated between distinct users versus those between non-distinct users.

Some guesswork and heuristics parsing on-chain transaction data is required in order to more accurately reflect the real economic activity occurring on a blockchain and compare it with the activity occurring on other blockchains or within different state economies.

CONCLUSION

While easily extracted, on-chain transaction data can be difficult to make sense of, when taken at face value. It can also be misleading at times if the data is not appropriately contextualized with off-chain data like a crypto asset's market price.

As a starting point for analysis, on-chain transaction count, total value transferred and velocity are three metrics that are useful for tracking changes in transaction activity for a crypto asset over time. The novelty of crypto assets is largely based on the way the technology of these assets enables transfers of value directly and without the oversight of a third party.

A non-zero value for any of these three metrics is an indication of platform utility. It cannot be assumed, however, what the predominant use case driving transaction activity for a crypto asset is.

As such, the next step for analysis that on-chain transaction data naturally points to is uncovering the off-chain circumstances and trends motivating users to transact with bitcoin, ether, and other crypto assets. The white paper and statements from core developers behind most crypto assets often gives some indication for intended use cases.

But the decentralized nature of crypto asset development leaves the practical application of these assets in the hands of its users, whoever and wherever they may be. This makes the analysis of crypto assets a highly iterative process that will continue to change as the user base for these assets also evolves.

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